Haonan Wu

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EDUCATION

The Pennsylvania State University (PSU), USA — Aug. 2023 - present
Ph.D. in Computer Science and Engineering
Research Topics: Theoretical and Algorithmic Bioinformatics
Supervisor: Prof. Paul Medvedev

Shandong University (SDU), China — Sep. 2020 - Jun. 2023
Master of Science in Operational Research and Cybernetics
GPA: 92.17/100
Relevant Coursework: Graphs and Digraphs, Combinatorial Optimization, Analysis and Design of Algorithms, Modern Functional Analysis, Elements of Modern Algebra, Mathematical Analysis

Shandong University (SDU), China — Sep. 2016 - Jun. 2020
Bachelor of Science in Biotechnology
GPA: 90.65/100
Relevant Coursework: Biochemistry, Cell Biology, Molecular Biology

WORK EXPERIENCE

The Pennsylvania State University — State College, USA
Research Assistant in Department of Computer Science and Engineering — Jan. 2024 – present
Supervisor: Prof. Paul Medvedev

Developing bioinformatics theories and algorithms.

RESEARCH INTERESTS

I am interested in developing theories, algorithms, and practical tools for bioinformatics, with the goal of addressing biological questions. My research is guided by two principles:

1. Developing methods grounded in rigorous mathematics and algorithms, or in clear mathematical intuition.
2. Building practically useful tools that incorporate effective heuristics and benefit the bioinformatics community.

Currently, my research focuses on repetitive genomic sequences from both theoretical and practical perspectives. My theoretical work focuses on the analysis of k-mer behavior under mutation and evolutionary processes, and estimation of the mutation rate. On the practical side, I develop algorithms specifically designed for centromeres, covering tasks such as annotation, variant calling, and sequence alignment.

PEER-REVIEWED PUBLICATIONS & PREPRINTS

Journals

- **Haonan Wu** and Paul Medvedev. The Gift of Novelty: Repeat-Robust k-mer-Based Estimators of Mutation Rates. *Bioinformatics*, 42(Supplement 1):btag234, 2026 (Proceedings Issue of ISMB 2026).
- **Haonan Wu**, Jiyun Han, Shizhuo Zhang, Gaojia Xin, Chaozhou Mou, and Juntao Liu. Spatom: A Graph Neural Network for Structure-Based Protein–Protein Interaction Site Prediction. *Briefings in Bioinformatics*, 24(6): bbad345, 2023. DOI: 10.1093/bib/bbad345

Conference Proceedings

- **Haonan Wu** and Paul Medvedev. The Gift of Novelty: Repeat-Robust Estimators of Substitution Rates. Accepted by the 34th Intelligent Systems for Molecular Biology (ISMB 2026).

- **Haonan Wu**, Antonio Blanca, and Paul Medvedev. A k-mer-Based Estimator of the Substitution Rate Between Repetitive Sequences. In *Proceedings of the 25th International Conference on Algorithms for Bioinformatics (WABI 2025)*, Leibniz International Proceedings in Informatics (LIPIcs), Vol. 344, pp. 20:1–20:20, 2025.

Preprints

- Roland Faure, Adrita Hossain Nakshi, Md Hasin Abrar, Rayan Chikhi, Teo Lemane, Mohsen Taheri, Stephanie Won, **Haonan Wu**, David Koslicki, Paul Medvedev. "Sketching the SRA: experiment-wide similarity search across 4.8 million metagenomes." *bioRxiv* 2026.10.07.757297; doi: https://doi.org/10.64898/2026.10.07.757297

RESEARCH EXPERIENCE

Repeat-robust estimators under different settings 2026
supervised by Dr. Paul Medvedev

- Developed three k-mer-based estimators robust to repetitive genomic regions, applicable under varying levels of count information (presence-only, one-sided counts, or full counts).
- Demonstrated state-of-the-art performance within each setting, with the strongest model outperforming existing methods on highly repetitive alpha satellite sequences.
- Code available at github.com/medvedevgroup/Accurate_repeat-aware_kmer_based_estimator

Repeat-aware substitution rate estimation for repetitive sequences 2025
supervised by Dr. Paul Medvedev

- Proposed a k-mer-based estimator that relaxes the non-repetitive assumption, enabling accurate mutation rate estimation in repetitive genomes.
- Derived theoretical bounds on estimator bias and validated robustness on centromeric higher-order repeat datasets.
- Code available at github.com/medvedevgroup/Repeat-Aware_Substitution_Rate_Estimator

Spatom: structure-based protein–protein interaction site prediction 2023
supervised by Dr. Juntao Liu

- Developed a graph neural network framework modeling protein structures as weighted digraphs to capture spatial residue interactions.
- Designed a hybrid architecture combining digraph convolution and attention mechanisms, achieving state-of-the-art performance on benchmark protein complexes.
- Code available at github.com/Wu-Haonan/Spatom
- Web server at http://liulab.top/Spatom/server

ACADEMIC SERVICE

- Reviewer: Bioinformatics; Heliyon; ACM-BCB 2025–2026 (primary reviewer); RECOMB 2025–2026 (sub-reviewer); ISMB 2025–2026 (sub-reviewer).
- Seminar Coordinator for the CSE Bioinformatics Joint-Group Meetings, Penn State (Spring 2026).

MENTORSHIP & ADVISING

Undergraduate Research Mentor

- Kathryn Ellen Silverberg, Pennsylvania State University April. 2026 - present
  Schreyer Honors Program thesis: Fast Computation of Hamming Distance Histograms for k-mers

Peer Mentorship (PhD Applications)
Advising on PhD applications, including research statements and application strategy.

- Anxuan Han 2024
  admitted to PhD program at the University of Adelaide
- Yangyang Liu 2025
  admitted to PhD program at the University of Southern California

TEACHING EXPERIENCE

Instructor

- Deep-learning Short Course, Shandong University Winter 2022
  - Designed and delivered a short course on deep learning fundamentals and practical implementation with PyTorch.
  - Course materials at github.com/Wu-Haonan/Deep_learning_short_course

Undergraduate Tutor ("Dolphin Program")
Tutoring on course material, supporting students' transition to college life, and offering guidance on academic planning.

- Inorganic and Analytical Chemistry, School of Marine, SDU, China Fall 2018
- Probability and Mathematical Statistics II, School of Marine, SDU, China Spring 2019

TALKS & POSTER

Talks

- Title: The Gift of Novelty:
  Repeat-robust k-mer-based estimators of mutation rates
  ISMB 2026, Washington, DC, USA Jul. 2026

- Title: Don't Repeat 'No Repeats':
  A k-mer-based estimator of the substitution rate between repetitive sequences
  Genome Informatics 2025, Cold Spring Harbor Laboratory, NY, USA Nov. 5-8 2025

- Title: Don't Repeat 'No Repeats':
  A k-mer-based estimator of the substitution rate between repetitive sequences
  WABI 2025, University of Maryland, MD, USA Aug. 20 2025

- Topic: Applications of Digraph in bioinformatics and computational biology
  Graphs and Digraphs, School of Mathematics, SDU, China Spring 2022

- Topic: Introduction of Single Molecular Sequence and assembly algorithms for long-reads
  Sequence and Assembly Seminar, School of Mathematics and Statistics, SDU, China Fall 2019

Posters

- A k-mer-based estimator of the substitution rate between repetitive sequences
  WABI 2025, University of Maryland, MD, USA (Link)
- A k-mer-based estimator of the substitution rate between repetitive sequences
  Rao Prize Conference 2025, Penn State University, PA, USA (Link)

AWARDS & FELLOWSHIPS

- ISMB Conference Fellowship, awarded by the International Society for Computational Biology (ISCB), 2026.

OTHER EXPERIENCES

Explored the possible reason for abundance of oxides on the Martian surface Jun. 2017 - Oct. 2018
Planetary Lab, Space Science School, SDU with Dr. Zhongchen Wu

SKILLS

- **Programming:** Python, MATLAB, C++, experience programming in Linux server
- **Package:** Pytorch, scikit-learn

- **Scripting Language:** LaTeX
- **Language:** Chinese (native speaker), English (working language)

REFERENCES

Prof. Paul Medvedev
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Prof. Antonio Blanca
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